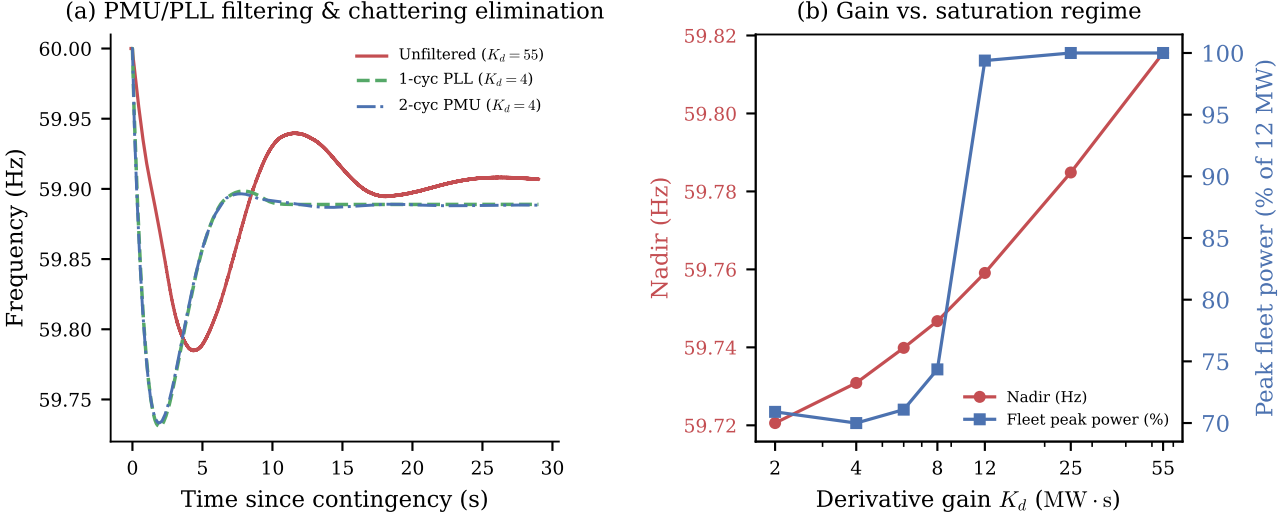

(a) PMU/PLL filtering & chattering elimination
Unfiltered ($K_d = 55$)
1-cyc PLL ($K_d = 4$)
2-cyc PMU ($K_d = 4$)
Frequency (Hz)
60.00
59.95
59.90
59.85
59.80
59.75
0
5
10
15
20
25
30
Time since contingency (s)
(b) Gain vs. saturation regime
Nadir (Hz)
59.82
59.80
59.78
59.76
59.74
59.72
Peak fleet power (% of 12 MW)
100
95
90
85
80
75
70
Nadir (Hz)
Fleet peak power (%)
2
4
8
12
25
55
Derivative gain K_d (MW · s)